UF Herbert Wertheim College of Engineering
UNIVERSITY of FLORIDA

Lecture 5G – Visual Search & Signal Detection Theory II

EIN 3241: Human Factors and Ergonomics
Spring 2026

March 13, 2026

Dr. Jad A. Atweh

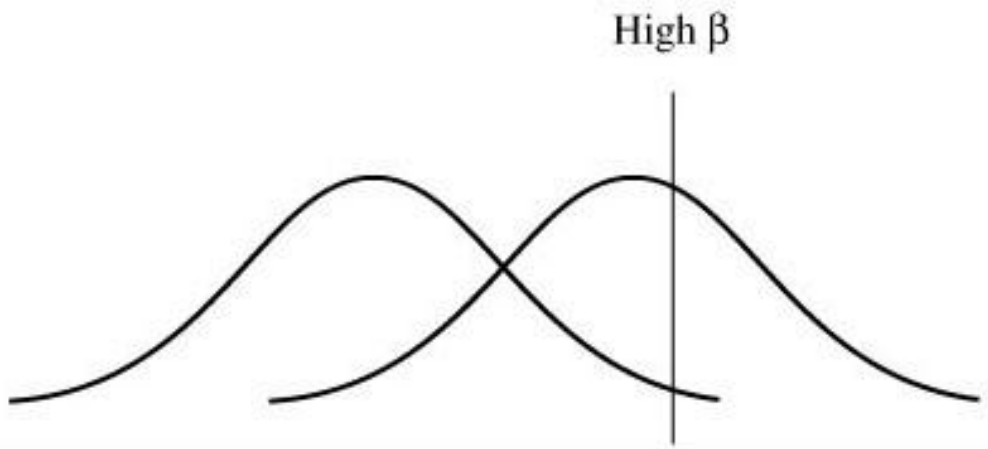
Admin Issues

- Research Study Last Reminder!
- Group project assignment (by Sunday night)
- Progress Report (2%): Monday, March 30
- Final Report (9%): April 25
- Presentations (9%): April 17, 20, and 22 (slots will be determined later).
- Previous Projects

Decision pattern & sensitivity

Probabilities based on observations. Need to ensure sample sizes are consistent across event types

	Signal	Noise
Yes	0.30	0.05
No	0.70	0.95



Use z score and normal curve ordinate tables to solve

Z(H) = Z(0.3)

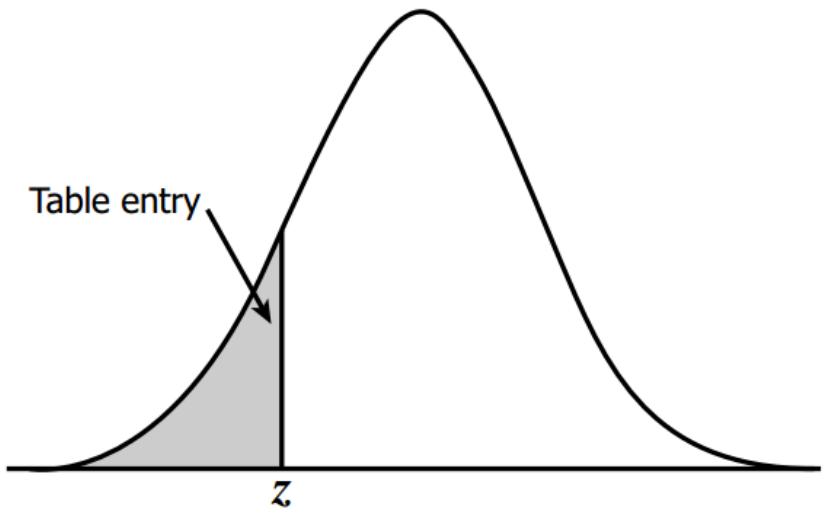
Ordinate(Z(H)) =

Z(FA) = Z(0.05)

Ordinate(Z(FA)) =

β = O(H) / O(FA)
=

d' = Z(H) - Z(FA)
=



z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
–3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002
–3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0004	.0003
–3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	.0005	.0005
–3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.0008	.0007	.0007
–3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
–2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014
–2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
–2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
–2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
–2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
–2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
–2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
–2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
–2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
–2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
–1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
–1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
–1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
–1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
–1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
–1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0708	.0694	.0681
–1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
–1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
–1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
–1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
–0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
–0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
–0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
–0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
–0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
–0.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
–0.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
–0.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
–0.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
–0.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641

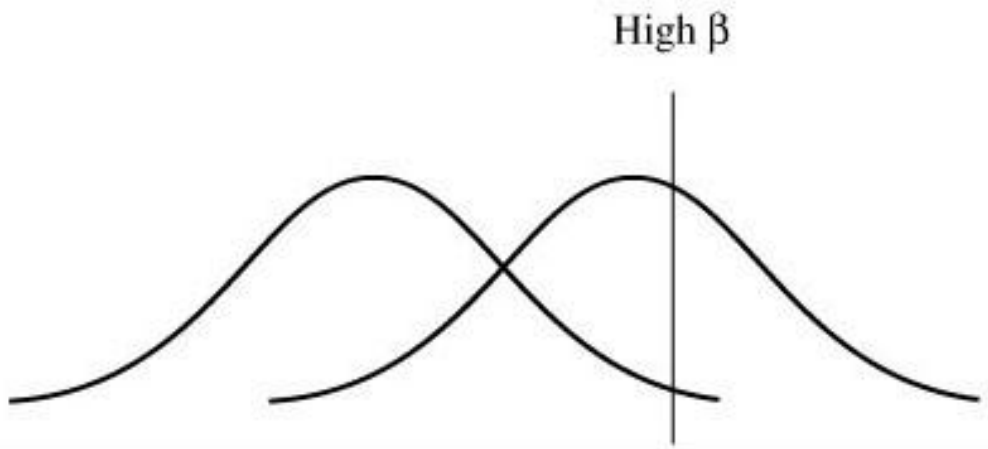
$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2}$, for $0 \le z \le 4.99, \phi(-z) = \phi(z)$

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.3989	0.3989	0.3989	0.3988	0.3986	0.3984	0.3982	0.3980	0.3977	0.3973
0.1	0.3970	0.3965	0.3961	0.3956	0.3951	0.3945	0.3939	0.3932	0.3925	0.3918
0.2	0.3910	0.3902	0.3894	0.3885	0.3876	0.3867	0.3857	0.3847	0.3836	0.3825
0.3	0.3814	0.3802	0.3790	0.3778	0.3765	0.3752	0.3739	0.3725	0.3712	0.3697
0.4	0.3683	0.3668	0.3653	0.3637	0.3621	0.3605	0.3589	0.3572	0.3555	0.3538
0.5	0.3521	0.3503	0.3485	0.3467	0.3448	0.3429	0.3410	0.3391	0.3372	0.3352
0.6	0.3332	0.3312	0.3292	0.3271	0.3251	0.3230	0.3209	0.3187	0.3166	0.3144
0.7	0.3123	0.3101	0.3079	0.3056	0.3034	0.3011	0.2989	0.2966	0.2943	0.2920
0.8	0.2897	0.2874	0.2850	0.2827	0.2803	0.2780	0.2756	0.2732	0.2709	0.2685
0.9	0.2661	0.2637	0.2613	0.2589	0.2565	0.2541	0.2516	0.2492	0.2468	0.2444
1.0	0.2420	0.2396	0.2371	0.2347	0.2323	0.2299	0.2275	0.2251	0.2227	0.2203
1.1	0.2179	0.2155	0.2131	0.2107	0.2083	0.2059	0.2036	0.2012	0.1989	0.1965
1.2	0.1942	0.1919	0.1895	0.1872	0.1849	0.1826	0.1804	0.1781	0.1758	0.1736
1.3	0.1714	0.1691	0.1669	0.1647	0.1626	0.1604	0.1582	0.1561	0.1539	0.1518
1.4	0.1497	0.1476	0.1456	0.1435	0.1415	0.1394	0.1374	0.1354	0.1334	0.1315
1.5	0.1295	0.1276	0.1257	0.1238	0.1219	0.1200	0.1182	0.1163	0.1145	0.1127
1.6	0.1109	0.1092	0.1074	0.1057	0.1040	0.1023	0.1006	0.09893	0.09728	0.09566
1.7	0.09405	0.09246	0.09089	0.08933	0.08780	0.08628	0.08478	0.08329	0.08183	0.08038
1.8	0.07895	0.07754	0.07614	0.07477	0.07341	0.07206	0.07074	0.06943	0.06814	0.06687
1.9	0.06562	0.06438	0.06316	0.06195	0.06077	0.05959	0.05844	0.05730	0.05618	0.05508
2.0	0.05399	0.05292	0.05186	0.05082	0.04980	0.04879	0.04780	0.04682	0.04586	0.04491
2.1	0.04398	0.04307	0.04217	0.04128	0.04041	0.03955	0.03871			

Decision pattern & sensitivity

Probabilities based on observations. Need to ensure sample sizes are consistent across event types

	Signal	Noise
Yes	0.30	0.05
No	0.70	0.95



Use z score and normal curve ordinate tables to solve

Z(H) = -0.524

Ordinate(Z(H)) = 0.3485

Z(FA) = -1.644

Ordinate(Z(FA)) = .1040

β = O(H) / O(FA)
= 0.3485 / .1040 = 3.35

d' = Z(H) - Z(FA)
= -0.524 – (-1.644) = 1.12 SD

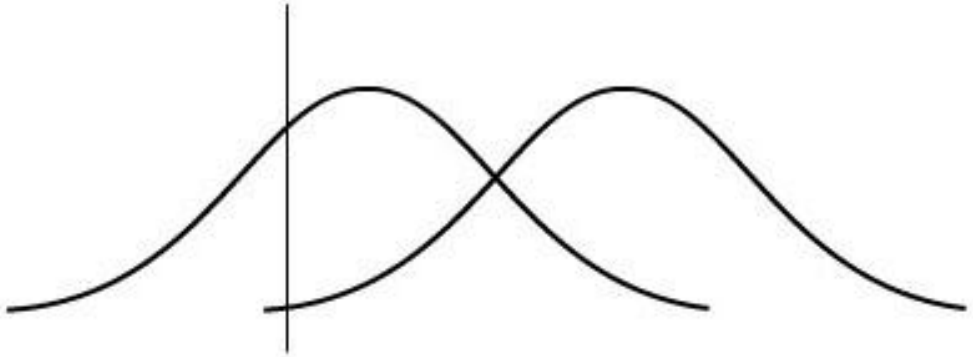
Standard Normal Scores

z	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
-0	.50000	.49601	.49202	.48803	.48405	.48006	.47608	.47210	.46812	.46414
-0.1	.46017	.45620	.45224	.44828	.44433	.44034	.43640	.43251	.42858	.42465
-0.2	.42074	.41683	.41294	.40905	.40517	.40129	.39743	.39358	.38974	.38591
-0.3	.38209	.37828	.37448	.37070	.36693	.36317	.35942	.35569	.35197	.34827
-0.4	.34458	.34090	.33724	.33360	.32997	.32636	.32276	.31918	.31561	.31207
-0.5	.30854	.30503	.30153	.29806	.29460	.29116	.28774	.28434	.28096	.27760
-0.6	.27425	.27093	.26763	.26435	.26109	.25785	.25463	.25143	.24825	.24510
-0.7	.24196	.23885	.23576	.23270	.22965	.22663	.22363	.22065	.21770	.21476
-0.8	.21186	.20897	.20611	.20327	.20045	.19766	.19489	.19215	.18943	.18673
-0.9	.18406	.18141	.17879	.17619	.17361	.17106	.16853	.16602	.16354	.16109
-1	.15866	.15625	.15386	.15151	.14917	.14686	.14457	.14231	.14007	.13786
-1.1	.13567	.13350	.13136	.12924	.12714	.12507	.12302	.12100	.11900	.11702
-1.2	.11507	.11314	.11123	.10935	.10749	.10565	.10383	.10204	.10027	.09853
-1.3	.09680	.09510	.09342	.09176	.09012	.08851	.08692	.08534	.08379	.08226
-1.4	.08076	.07927	.07780	.07636	.07493	.07353	.07215	.07078	.06944	.06811
-1.5	.06681	.06552	.06426	.06301	.06178	.06057	.05938	.05821	.05705	.05592
-1.6	.05480	.05370	.05262	.05155	.05050	.04947	.04846	.04746	.04648	.04551
-1.7	.04457	.04363	.04272	.04182	.04093	.04006	.03920	.03836	.03754	.03673
-1.8	.03593	.03515	.03438	.03362	.03288	.03216	.03144	.03074	.03005	.02938
-1.9	.02872	.02807	.02743	.02680	.02619	.02559	.02500	.02442	.02385	.02330
-2	.02275	.02222	.02169	.02118	.02068	.02018	.01970	.01923	.01876	.01831
-2.1	.01786	.01743	.01700	.01659	.01618	.01578	.01539	.01500	.01463	.01426
-2.2	.01390	.01355	.01321	.01287	.01255	.01222	.01191	.01160	.01130	.01101
-2.3	.01072	.01044	.01017	.00990	.00964	.00939	.00914	.00889	.00866	.00842
-2.4	.00820	.00798	.00776	.00755	.00734	.00714	.00695	.00676	.00657	.00639
-2.5	.00621	.00604	.00587	.00570	.00554	.00539	.00523	.00508	.00494	.00480
-2.6	.00466	.00453	.00440	.00427	.00415	.00402	.00391	.00379	.00368	.00357
-2.7	.00347	.00336	.00326	.00317	.00307	.00298	.00289	.00280	.00272	.00264
-2.8	.00256	.00248	.00240	.00233	.00226	.00219	.00212	.00205	.00199	.00193
-2.9	.00187	.00181	.00175	.00169	.00164	.00159	.00154	.00149	.00144	.00139
-3	.00135	.00131	.00126	.00122	.00118	.00114	.00111	.00107	.00104	.00100
-3.1	.00097	.00094	.00090	.00087	.00084	.00082	.00079	.00076	.00074	.00071
-3.2	.00069	.00066	.00064	.00062	.00060	.00058	.00056	.00054	.00052	.00050
-3.3	.00048	.00047	.00045	.00043	.00042	.00040	.00039	.00038	.00036	.00035
-3.4	.00034	.00032	.00031	.00030	.00029	.00028	.00027	.00026	.00025	.00024
-3.5	.00023	.00022	.00022	.00021	.00020	.00019	.00019	.00018	.00017	.00017
-3.6	.00016	.00015	.00015	.00014	.00014	.00013	.00013	.00012	.00012	.00011
-3.7	.00011	.00010	.00010	.00010	.00009	.00009	.00008	.00008	.00008	.00008
-3.8	.00007	.00007	.00007	.00006	.00006	.00006	.00006	.00005	.00005	.00005
-3.9	.00005	.00005	.00004	.00004	.00004	.00004	.00004	.00004	.00003	.00003
-4	.00003	.00003	.00003	.00003	.00003	.00003	.00002	.00002	.00002	.00002

Decision pattern & sensitivity

	Signal	Noise
Yes	0.95	0.70
No	0.05	0.30

Low β



Type of responder?

Z(H) =

Ordinate(H) =

Z(FA) =

Ordinate(FA) =

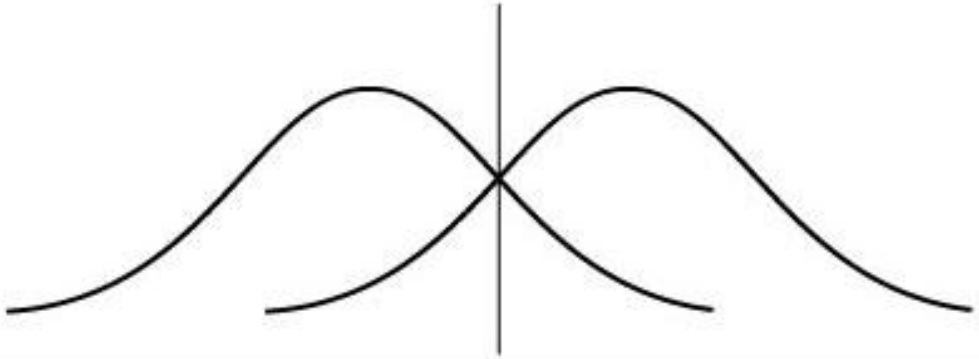
β =

d’ =

Decision pattern & sensitivity

	Signal	Noise
Yes	0.70	0.30
No	0.30	0.70

$\beta = 1$



Type of responder and level of sensitivity?

Z(H) =

Ordinate(H) =

Z(FA) =

Ordinate(FA) =

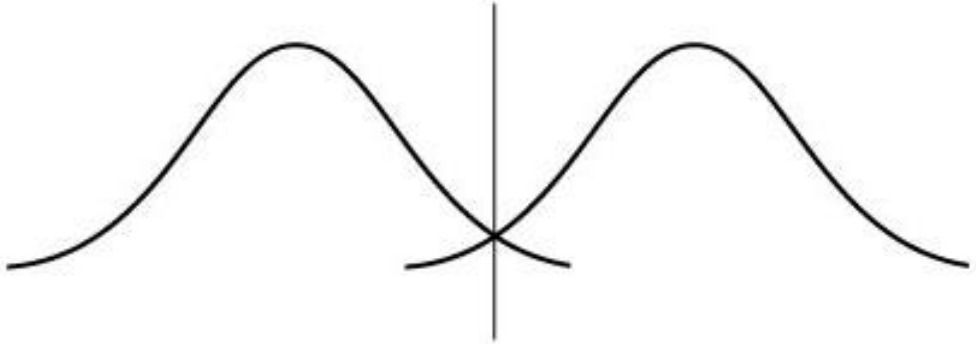
β =

d’ =

Decision pattern & sensitivity

	Signal	Noise
Yes	0.90	0.10
No	0.10	0.90

$\beta = 1$



Type of responder and level of sensitivity?

Z(H) =

Ordinate(H) =

Z(FA) =

Ordinate(FA) =

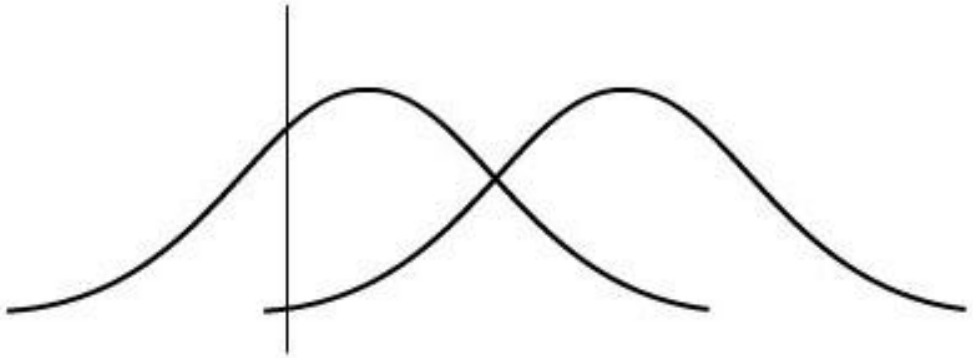
β =

d’ =

Decision pattern & sensitivity

	Signal	Noise
Yes	0.95	0.70
No	0.05	0.30

Low β



Type of responder?

Z(H) = 1.644

Ordinate(H) = 0.103

Z(FA) = 0.524

Ordinate(FA) = 0.348

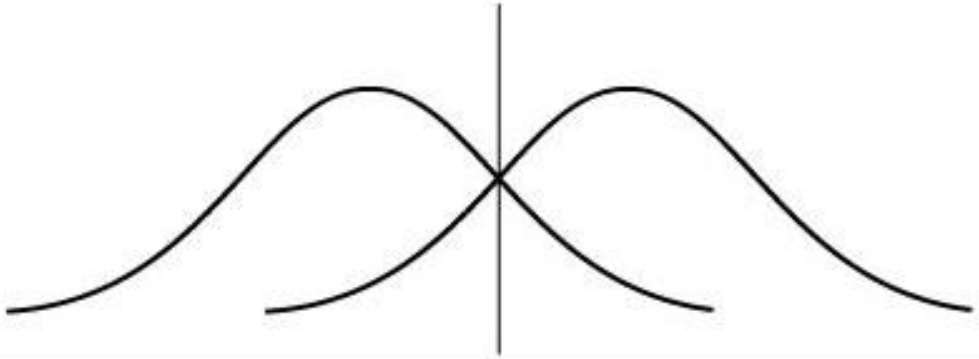
β = 0.103 / 0.348 = 0.297

d’ = 1.644 – 0.524 = 1.12

Decision pattern & sensitivity

	Signal	Noise
Yes	0.70	0.30
No	0.30	0.70

$\beta = 1$



Type of responder and level of sensitivity?

Z(H) = 0.524

Ordinate(H) = 0.348

Z(FA) = -0.524

Ordinate(FA) = 0.348

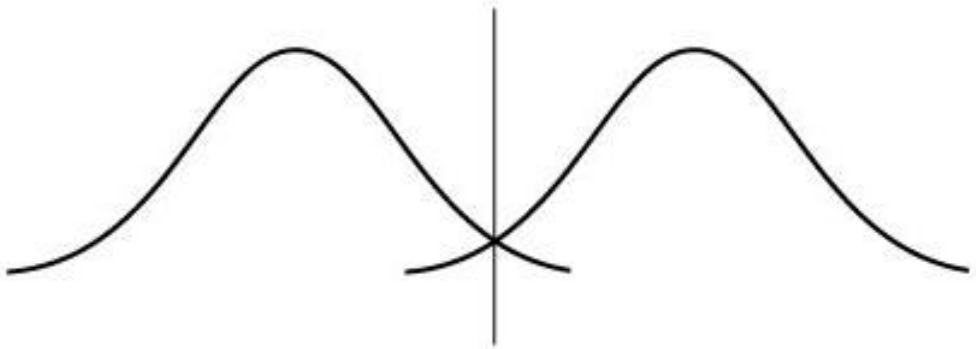
$\beta = 1$

d’ = 1.049

Decision pattern & sensitivity

	Signal	Noise
Yes	0.90	0.10
No	0.10	0.90

$\beta = 1$



Type of responder and level of sensitivity?

Z(H) = 1.282

Ordinate(H) = 0.175

Z(FA) = -1.282

Ordinate(FA) = 0.175

$\beta = 1$

d’ = 2.563

Applications of SDT

- Medical diagnosis
- Police identification line-ups
- Objects on a radar screen
- Discriminating smells [gas leak detection]
- Drug testing in personnel selection (fluctuations in sensitivity levels)
- Military detection (camouflaged objects)
- Use of night vision devices for object detection
- Emergency / fire alarm detection – perception of ambulances on roadway

More on response criterion…

- β can be manipulated by…
  - Probabilities of noise and signals
    - Many stimuli present in trials, you lower criterion.
    - Few stimuli present in trials, you raise criterion.
  - Payoffs associated with the SDT outcomes

- Optimal β can be defined based on probabilities alone, as:

$$\beta_{opt} = \frac{P(N)}{P(S)}$$

  - If $P(N) = P(S)$, then optimal Beta = 1.

Response criterion and payoffs

- Payoffs can also be considered in model of optimal β:
  - If subject receives \$100 for hit, but is only penalized \$1 for a FA, will produce low/risky criterion.
  - If operator saves company \$100 with correct rejection, but penalizes company \$1 for miss, will produce high/conservative criterion.
- Optimal β can be defined based outcome payoffs (alone), as:

$$\beta_{opt} = \frac{V(CR) + C(FA)}{V(H) + C(M)} = \frac{Outcome\ in\ event\ of\ noise}{Outcome\ in\ event\ of\ signal}$$

V(CR) = Value of correct rejection
C(FA) = Cost of false alarm
V(H) = Value of hit
C(M) = Cost of miss

Combining equations

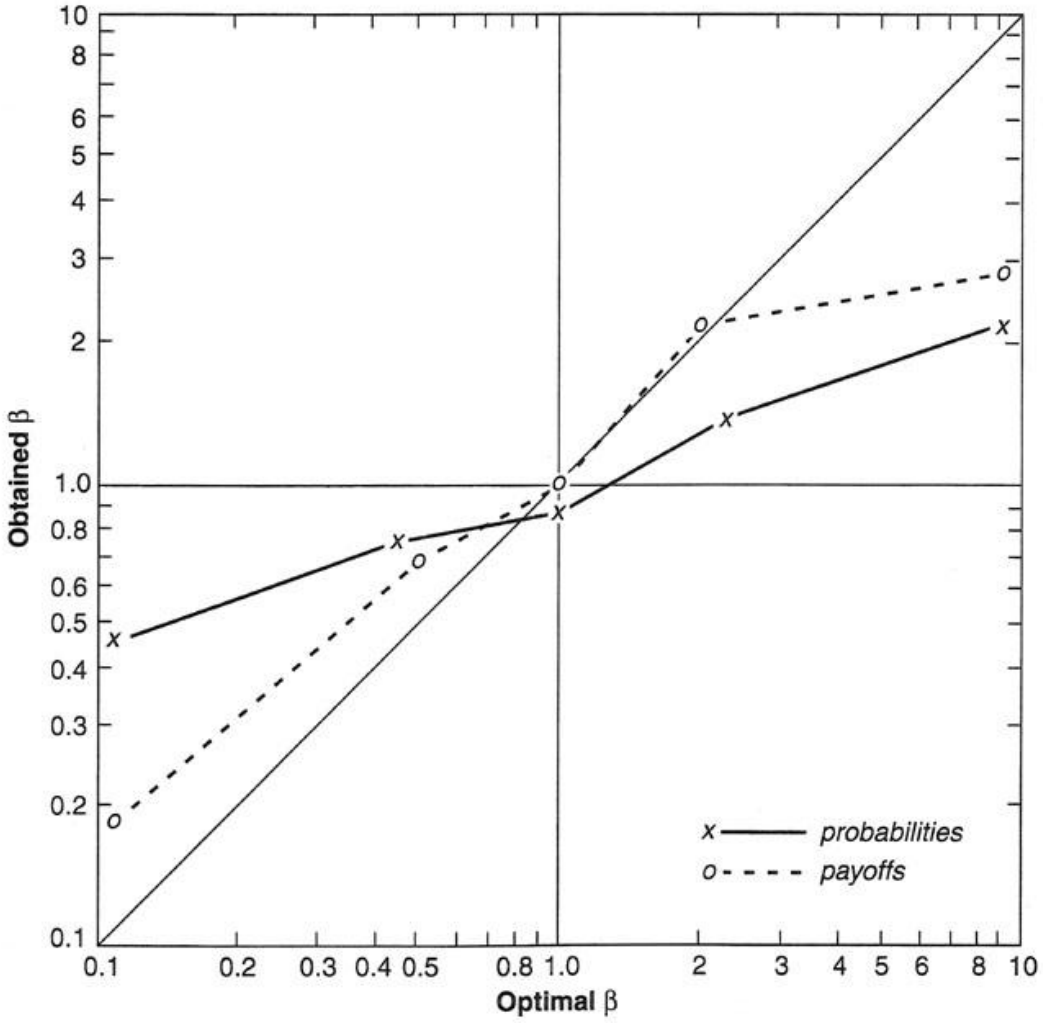
$$\beta_{opt} = \frac{P(N)}{P(S)} * \frac{V(CR) + C(FA)}{V(H) + C(M)} = \frac{Probability\ of\ Noise}{Probability\ of\ Signal} * \frac{Outcome\ in\ event\ of\ noise}{Outcome\ in\ event\ of\ signal}$$

- If you know probabilities and payoffs for SDT task, you can calculate optimal β.
- You can also compare actual operator performance (observed β) with optimal β.
  - People don't update or optimize β, based on changes in event probabilities, as often as they should.
    - Called this "**sluggish β**"!

Sluggish beta

- Optimal β values for task on X-axis.
- Actual β values on Y-axis.
- Optimal performance occurs when obtained β = optimal β.

  (Positive diagonal)
- If manipulate probabilities (x) and payoffs (o) independently, change in β is slower for probabilities.

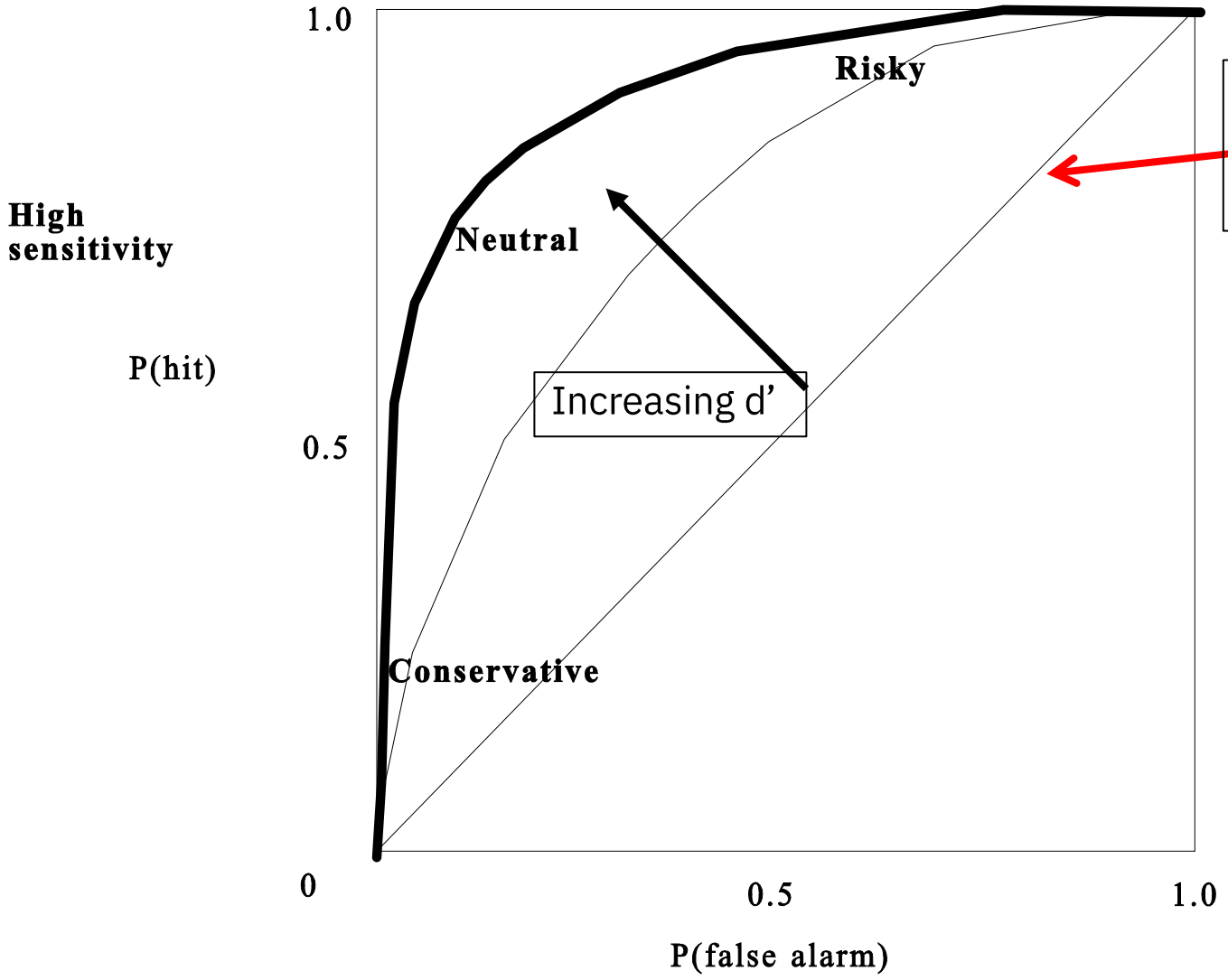


Receiver Operating Characteristic (ROC) Curve

- Multiple outcomes of signal detection analysis represented in same plot to characterize receiver behavior – ROC curve
  - Plot of P(H) against P(FA)
  - Recall that…

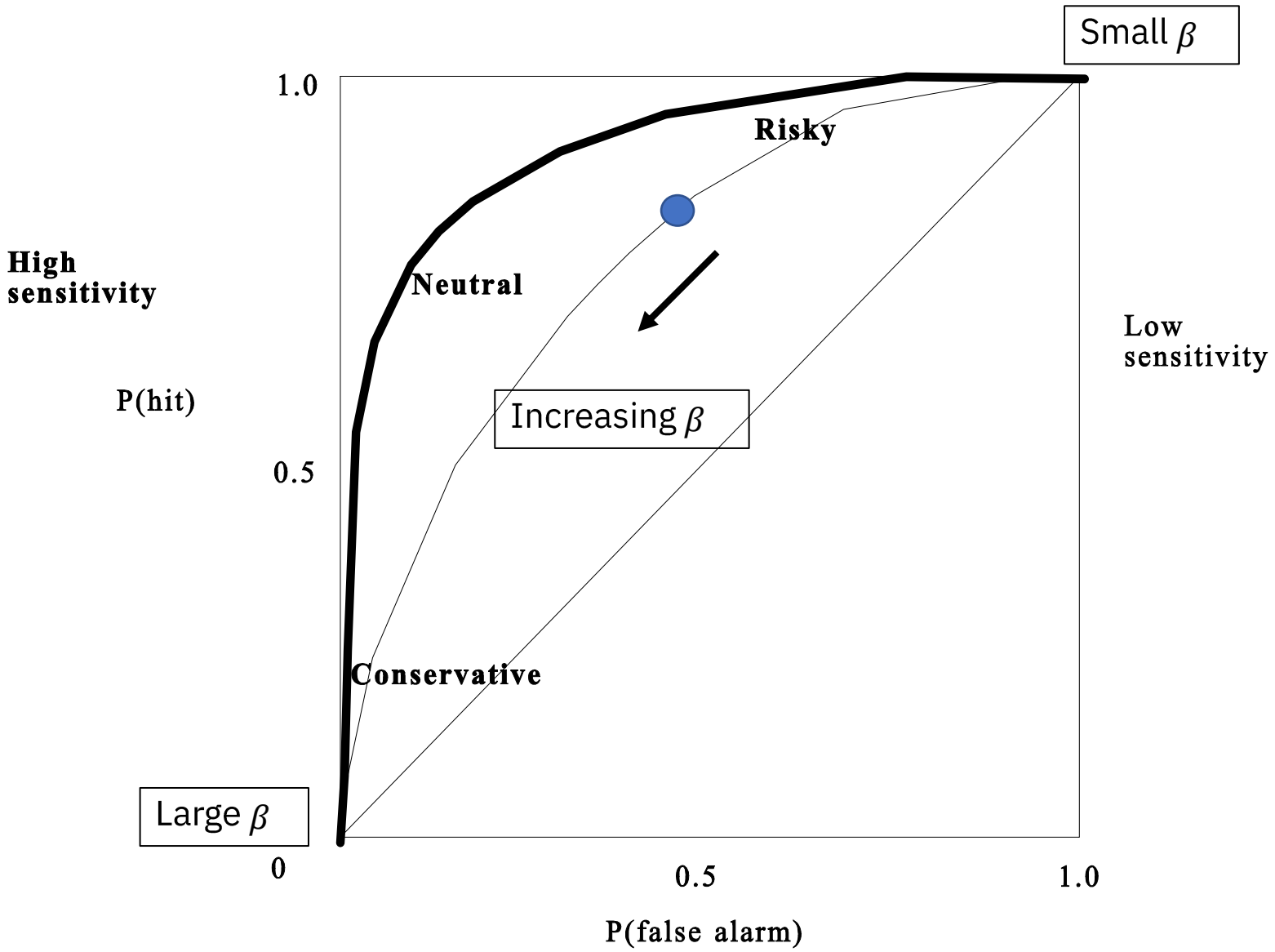
    $P(M) = 1-P(H)$

    $P(CR) = 1-P(FA)$
- Shows how observer adjusts response criterion (β) under different task circumstances (e.g., event probabilities):
  - Assumes signal strength and sensitivity (d') are constant.
  - Observations fall along bowed curve in plot.
- Also known as an iso-sensitivity curve.



No better than "blind" guessing

d' = 0

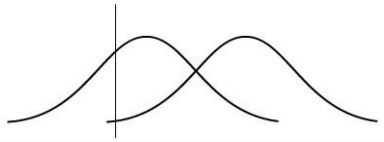

Small β
1.0
Risky
High sensitivity
Neutral
Low sensitivity
P(hit)
Increasing β
0.5
Conservative
Large β
0
0.5
1.0
P(false alarm)

ROC curves from SDT outcomes

H	FA
M	CR

	Signal	Noise
Yes	0.95	0.70
No	0.05	0.30

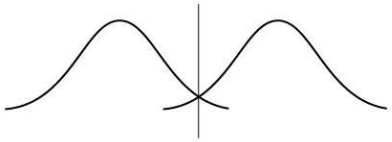
Low β



$d' = 1.12$

	Signal	Noise
Yes	0.90	0.10
No	0.10	0.90

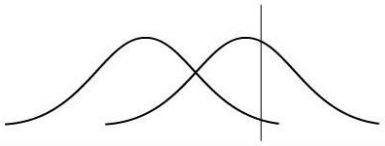
β = 1



$d' = 2.57$

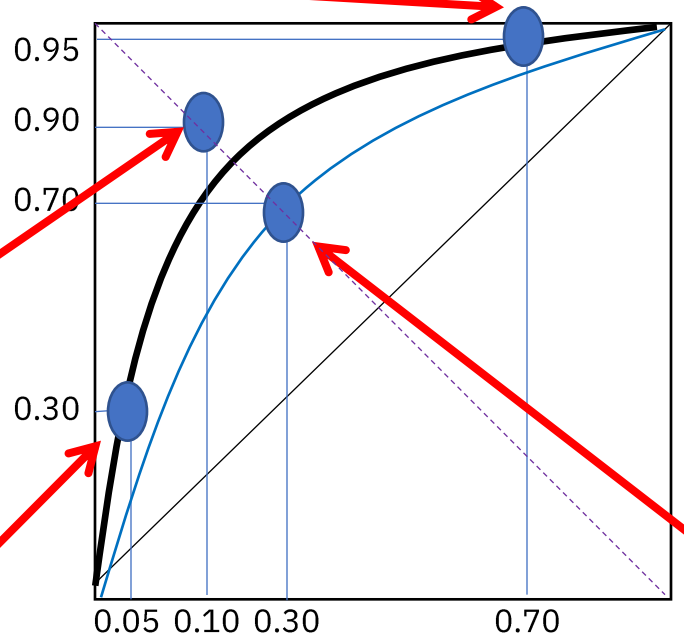
	Signal	Noise
Yes	0.30	0.05
No	0.70	0.95

High β



$d' = 1.12$

Probability of Hit

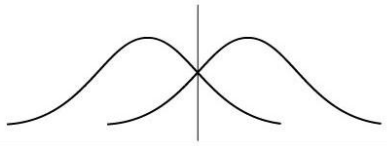


Probability of False Alarm

$d' = 1.05$

	Signal	Noise
Yes	0.70	0.30
No	0.30	0.70

β = 1



Calculate SDT outcomes

- Calculate $P(H)$ and $P(FA)$ with these data.

Response	"Signal"	"Noise"
"Yes"	14 (H)	6 (FA)
"No"	6 (M)	14 (CR)
Total observations (T)	20 (S)	20 (N)

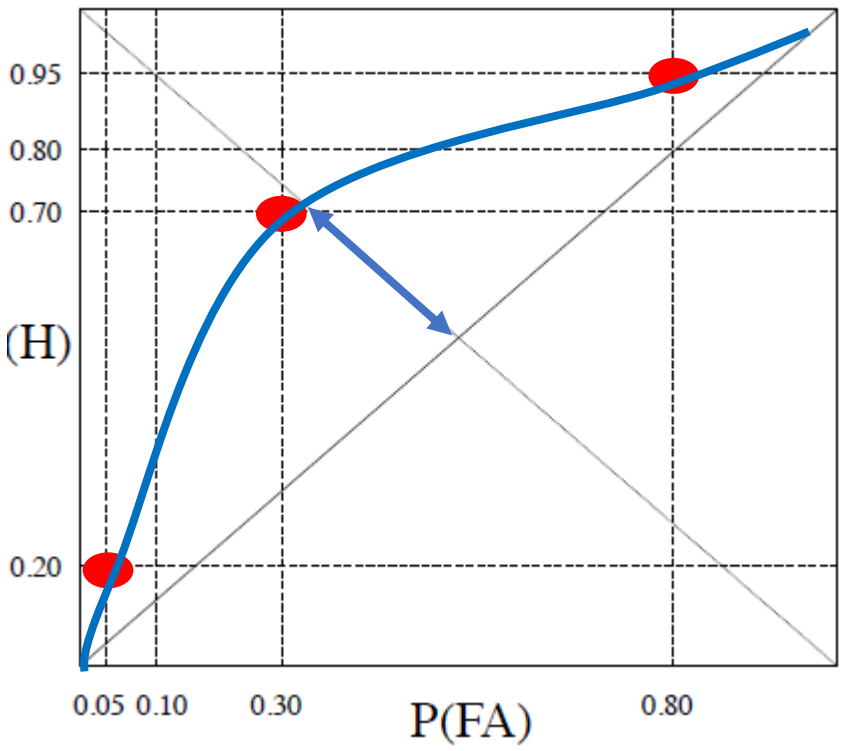
- $P(H)$ = 14 / 20 ≈ 0.7
- $P(M)$ = 6 / 20 ≈ 0.3
- $P(FA)$ = 6 / 20 ≈ 0.3
- $P(CR)$ = 14 / 20 ≈ 0.7

Plot ROC curve

- Plot $P(H)$ and $P(FA)$ values with single dot.

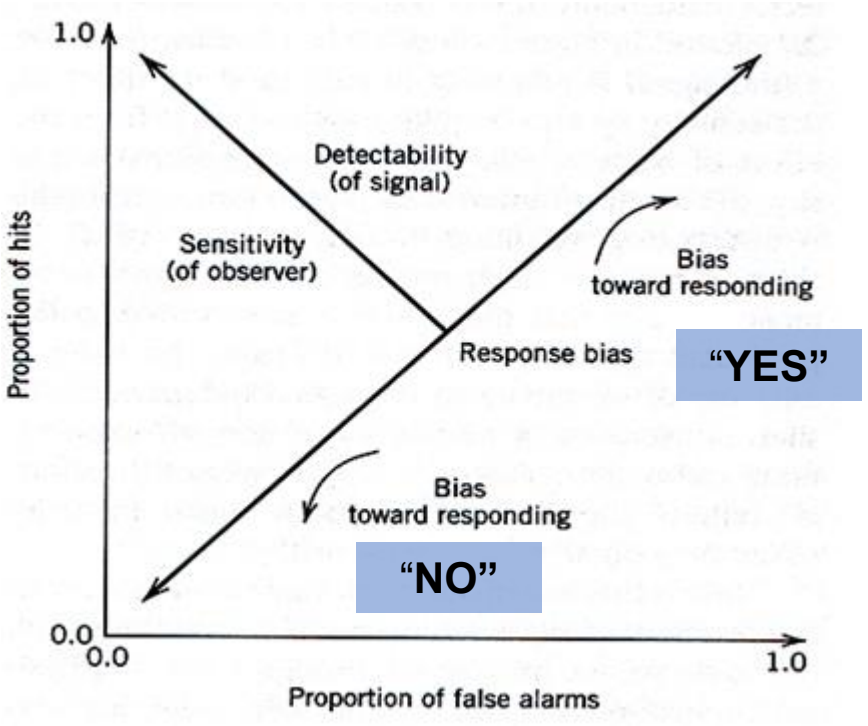
Response	"Signal"	"Noise"
"Yes"	0.2 (H)	0.05 (FA)
"No"	0.8 (M)	0.95 (CR)
Total observations (T)	20 (S)	20 (N)

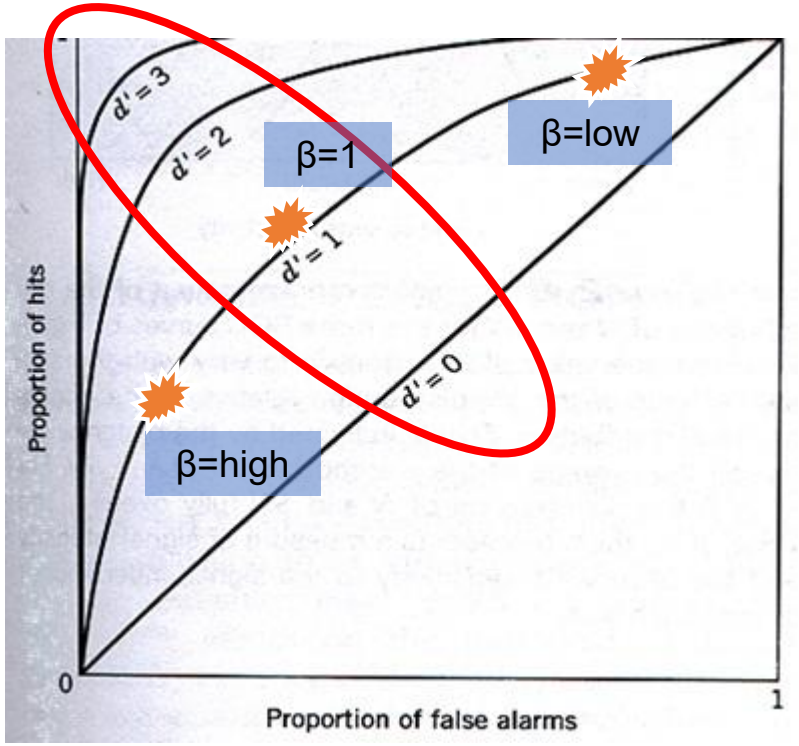
Response	"Signal"	"Noise"
"Yes"	0.95 (H)	0.8 (FA)
"No"	0.05 (M)	0.2 (CR)
Total observations (T)	20 (S)	20 (N)



- Connect dots to create ROC curve for user.
- Always represent β = 1 line for reference.
- Positive diagonal used as reference for judging level of sensitivity (d').
  - Graphically represented as distance between "chance" line and ROC curve at point of intersection with "unbias" response line.

Review of features and meaning





Human factors implications

- Why do we use ROC curve?
  - Provides representation of how observer changes decision criterion under different task circumstances.
  - What task factors may have influenced β shifts in our example data.
    - probabilities
    - payoffs
- For system and task designers, ROC gives us insight into:
  - how to define payoffs and penalties for optimal performance;
  - how to manipulate signal frequency (if possible) for best possible β (relative to optimum).

Human factors implications

- **Sensitivity loss:**
  1. Refresh with examples of the target (pictures)
     - Reduces memory load by providing an aid
  2. Training to improve sensitivity
     - Automaticity can reduce workload and make it easier to distinguish between targets and distractors
- **Response bias shift:**
  1. Feedback on success / failure
     - Helps people get a better estimate of the probability of noise and signals
  2. Introduce false signals
     - Also works as a memory aid
  3. Allow use of confidence levels in decision
     - Provide ratings of confidence rather than yes/no

UF

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LEADING THE CHARGE, CHARGING AHEAD